

Are current demand response baseline designs suitable for electric vehicles? Policy insights from the independent aggregation business model

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HIGHLIGHTS

- Baseline design selection significantly affects charging cost savings.
- Electric vehicle aggregators can manipulate certain baselines to boost profits.
- Policy makers must ensure fair competition among flexibility providers.

ARTICLE INFO

Keywords:

Balancing market
Baseline
Demand response
Electric vehicles
Independent aggregator
Smart charging

ABSTRACT

Demand response baselines are pivotal for accurately quantifying market-based flexibility provided by electric loads. Although they have been successfully applied to a variety of curtailable loads, their effectiveness for plug-in electric vehicles remains under question. This paper assesses the performance of current baseline designs for flexible electric vehicle fleets by examining the business case of an independent electric vehicle aggregator that participates in the electricity balancing market. A comprehensive market participation framework is developed to evaluate whether independent electric vehicle aggregators can strategically manipulate common baseline designs to maximize revenues through leveraged flexibility. Four different baseline designs are evaluated using year-long simulations at 15-min intervals, employing two real-world electric vehicle fleet datasets and actual data from two European wholesale electricity markets. The results shed light on the aggregator's revenues and flexibility provision, revealing that in two of the four baseline designs, the aggregator can manipulate baselines to significantly increase revenues by providing excessively high flexibility volumes, thereby calling the fairness of these designs into question. Our findings are valuable for policymakers and regulators, whose timely interventions are critical to prevent unfair competition among flexibility providers and ensure the long-term viability of the demand response market.

1. Introduction

Modern power systems are characterized by a high share of intermittent and variable renewable energy sources. The efficient and reliable operation of such systems requires adequate flexibility to maintain power balance. Demand Response (DR) is widely considered as a key provider of such flexibility, through the control of flexible electric loads, i.e., loads that can be curtailed or shifted across different time frames [1]. Because individual loads often have limited capacities, aggregation [2,3], which implies combining many such loads into a large portfolio,

enables aggregators to offer substantial flexibility to system operators through market-based mechanisms, earning revenues that are shared with participating consumers as an incentive to join DR programs [4].

Well-designed flexibility markets require a reliable measurement of the delivered flexibility. This measurement, known as the “baseline”, estimates the electricity that would have been consumed in the absence of a DR event [5]. Since baselines involve assumptions about “what would have happened”, robust baselining methods must balance the following principles to establish a fair playing field for all participants [6]:

Abbreviations: BM, Balancing Market; DR, Demand Response; DAM, Day Ahead Market; EBGL, Commission Regulation (EU) 2017/2195 on electricity balancing; EV, Electric Vehicle; GCT, Gate Closure Time; IEVA, Independent Electric Vehicle Aggregator; mFRR, Frequency Restoration Reserve with manual activation; MBMA, Meter Before – Meter After, baseline design; MTU, Market Time Unit, i.e., 15-min period; TSO, Transmission System Operator.

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<https://doi.org/10.1016/j.apenergy.2025.126281>

Received 17 February 2025; Received in revised form 21 May 2025; Accepted 1 June 2025

Available online 9 June 2025

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- **Accuracy**, for ensuring that the flexibility provided is neither overestimated for the buyer, nor underestimated for the provider.
- **Fairness (or Integrity [7])**, for limiting the opportunity to “manipulate” or “game” baselines, while ensuring consistency for consumers and a level playing field for market participants.
- **Simplicity**, for making baselines straightforward to implement, replicate, understand and verify from available data, which increases trust among participants.
- **Interoperability**, for enabling easy access to different flexibility services, and a common framework for coordinating this.

The policy implications of inaccurate and unfair baseline designs are significant, as they can result in erroneous credit/debit allocations among market participants, unfair competition among flexibility resources and loss of stakeholder trust – factors that could threaten DR market viability. Several studies have assessed the efficiency of common baseline designs using large, multi-year datasets that cover a variety of assets or customer segments. For instance, [8] evaluated a common baseline design based on the consumption of the ten most recent qualifying days, showing that it appears accurate only for large, industrial or commercial, weather-insensitive loads. The authors in [9] assessed the accuracy of eleven variations of common baseline designs across different commercial and industrial customers over a two-year project, concluding that hybrid baselines seem to be the most attractive option for customer participation. In [10] the accuracy of standard baselines and their adjusted versions was evaluated on a commercial load, demonstrating improved performance for the adjusted baselines, while [11] evaluated multiple baseline methodologies for residential loads, finding that historical and regression-based techniques can provide precise estimations. Additional studies on baseline designs and their performance in terms of accuracy and cost are discussed in a comprehensive review of baseline designs applied in European H2020 projects [12], which highlights that the choice of baseline design involves trade-offs among complexity, accuracy, and cost. A common feature of these studies is that they evaluate baseline designs in terms of accuracy, but not fairness. In addition, all evaluations are ex-post, failing to evaluate the strategic behavior a DR provider might adopt during the DR provision.

Beyond conventional flexible loads, plug-in Electric Vehicles (EVs) represent an emerging resource with considerable flexibility potential. In scenarios such as overnight charging or workplace charging, EVs often remain plugged in for several hours after reaching full charge, until the driver departs. Instead of immediate full charge upon connection, charging can be shifted to a later time, provided it is completed before the desired departure time. This approach ensures the same driving experience for EV users while potentially reducing their charging costs. The coordinated management of this flexible charging process is commonly referred to as smart charging [13]. Unlike many types of commercial and/or industrial loads where flexibility potential is simpler to quantify, the flexibility assessment of EVs is more challenging due to their high load variability, frequent activation, and the requirement for energy recovery [6]. The aggregation of large EV fleets helps mitigate uncertainties and enables the provision of a more controllable and reliable flexibility resource to the market. This is because individual EV charging behaviours tend to be variable and uncertain; however, when many such charging trajectories are aggregated, especially when they are imperfectly correlated, the fluctuations tend to average out. As a result, the total portfolio exhibits smoother and more predictable charging behavior [14,15]. Despite the enormous potential of smart charging, which will be an essential operation to avoid grid congestion problems [3], current baseline designs have not yet been thoroughly evaluated for their effectiveness on EV flexibility provision, particularly regarding accuracy and fairness. The only relevant work is [15], which evaluated the accuracy of one common baseline design on EV fleets using ex-post assessment, questioning its performance. However, a comprehensive evaluation of the fairness criterion necessitates a

comparison among multiple designs within a suitable market regime to highlight the potential for manipulating baseline levels.

Flexibility provision from EVs can be managed either by electricity suppliers (hereafter, suppliers) who seek to expand their core activities, or by independent entities focusing solely on the flexible provision through aggregated portfolios. The first approach is known as the “integrated” model and the second is known as the “independent aggregator” model [16]. Although the suppliers are naturally well-positioned to access, collect and analyse the flexibility potential of their customers, they have been slow in the development of market-based DR services [17] as this role is fundamentally different from their core business, i.e., the energy procurement to the end consumers from the wholesale market [18]. Thus, the independent DR aggregator emerges as an alternative viable entity for easier access of consumers in DR programs. Its importance is highlighted in recent European legislation, where Art. 17 of the Commission Regulation [19] established the formal recognition of the independent aggregator as a distinct energy market participant. The independent aggregators do not supply energy to the consumers and as the Directive foresees, they are able to access the flexibility markets without suppliers’ prior consent. This concept is a cornerstone towards the development of a more flexible and competitive energy market.

In scientific literature, numerous flexibility market participation frameworks have been presented with the scope to optimize market-based flexibility provision from EV fleets. However, the common approach is the “integrated” model, in which the supplier serves as the flexibility provider and makes optimal decisions for both energy procurement and flexibility provision. Frameworks employing bidding strategies for Day-Ahead Market participation are discussed in [20–24], while those addressing participation in both Day-Ahead and Ancillary Services are found in [25–31]. Despite numerous publications, no study has yet evaluated the effectiveness of current DR baseline designs. Moreover, a thorough assessment of baseline designs effectiveness must be based on the “independent aggregator” model. In this model, the aggregator focuses solely on maximizing revenues from providing EV flexibility in the most suitable flexibility market, without concerns about energy supply or energy imbalance costs that continue to challenge suppliers. This clear profit-driven scope makes the independent aggregator model suitable for revealing potential opportunities and threats for DR manipulation.

Towards this direction, in this work we develop a comprehensive optimization framework based on mixed integer linear programming for an independent EV aggregator. This framework models how EV fleets can optimally provide flexibility to system operators by participating in an existing European balancing market under real-world market rules. Using this framework, we address the following research question:

- Are current demand response baseline designs suitable for electric vehicles?

We first examine the question from a business perspective:

- Is the independent EV aggregation model viable under current DR baseline designs and market conditions?

To answer the question, we apply the developed framework under four real-world DR baseline designs. Multiple, one year-long simulations are carried out, with 15-min granularity, using two real-world EV fleet datasets and actual clearing prices from two European wholesale electricity markets with different imbalance mechanism designs. The seasonal and yearly numeric results provide novel insights into:

- I. The independent aggregator’s revenues and flexibility provision
- II. The impact of independent EV aggregation on the supplier costs for providing charging energy
- III. The charging cost saving potential for the EV users

We then can address the research question from a policy perspective:

- Can independent EV aggregators strategically manipulate common baseline designs to maximize revenues though leveraged flexibility?

Our results show that in two out of the four baseline designs the EV aggregator can manipulate baselines to increase revenues significantly by providing very high flexibility volumes, questioning the fairness criterion. This opportunity has serious policy implications which are analysed at the end of the paper.

The body of this paper is organized as follows: [Section 2](#) provides an overview of the electricity market framework, while [Section 3](#) includes the market participation framework. [Section 4](#) presents the case study design while [Section 5](#) includes the analysis and discussion of the case study results. Lastly, [Section 6](#) includes the conclusions and the policy implications of the conducted research.

2. Electricity market framework

This section includes the presentation of the European energy market framework.

2.1. European electricity balancing market

The organization of short-term electricity trading in Europe involves a series of successive markets, including the forward market, the day-ahead and the intra-day markets. The Balancing Market (BM) is the main European ancillary market for market-based active power balancing of the power system. The technical and operational aspects of BMs in Europe are governed by the provisions and specific rules outlined in the “Commission Regulation (EU) 2017/2195 of 23 November 2017 establishing a guideline on electricity balancing (EBGL)” [32].

EBGL establishes a robust framework for the European BM target model, outlining tasks and timelines for the development of European platforms dedicated to exchange BM products and ensuring fair, transparent and non-discriminatory participation of all technologies under common rules. It identifies various balancing products, such as Frequency Containment Reserve, Frequency Restoration Reserve with automatic activation, Frequency Restoration Reserve with manual activation (mFRR), and Replacement Reserve, each addressing different flexibility needs with varying response times. EBGL assumes that reserve capacity is procured through prior market mechanisms, and that upward/downward balancing energy offers are activated close to real-time

for balancing procurement. Capacity reservation allows Transmission System Operators (TSOs) to handle any abnormal situations in real-time operation by activating the reserved capacity, when necessary, instructing the balancing service providers to deliver upward/downward balancing energy based on their respective balancing energy offers submitted to the BM. Moreover, in this paper the TSO is assumed to take proactive measures to address real-time imbalances by relying on forecasted imbalances for the upcoming Market Time Unit(s) (MTUs) to clear the BM. The quantification of balancing energy needs for the next MTU follows the typical BM design as proposed by ENTSO-E [33,34].

This paper focuses on the mFRR provision. mFRR is deemed a suitable balancing product for Independent Electric Vehicle Aggregators (IEVAs) due to the attractive revenue potential in the mFRR energy market. Additionally, while capacity market participation for flexible resources such as DR and renewable portfolios is mandatory for some products, it is often voluntary for the mFRR capacity market in many countries [32,35,36]. Bidding in capacity markets typically occurs many hours before real-time, which poses significant challenges due to the requirement for activation of the reserved capacity in real-time upon request. The risk for the IEVAs is high, given the potential for unexpected changes in EV fleet behavior. Therefore, by selecting the mFRR participation, IEVAs can bypass the mFRR capacity market and focus on balancing energy offers submission closer to real-time, which provides greater control of available flexibility close to real-time. Finally, changes in EV charging rates for mFRR participation are required within a 15-min timeframe, while faster reserve types require adjustments every few seconds, making them more challenging. Currently, many European countries set the minimum bid size for mFRR participation at 1 MW. However, there is an ongoing discussion among system operators and regulators about lowering this threshold to enable wider participation and unlock the flexibility potential of distributed energy resources. Indeed, several of the world’s more advanced balancing markets have already reduced the minimum bid size to 0.1 MW [37].

In order to effectively design the optimal BM participation for the IEVA, the following assumptions, which reflect the current European BM design and are depicted in [Fig. 1](#), are made: i) the BM Gate Closure Time (BM GCT) for offers submission by the IEVA is defined 25 min before the beginning of the MTU, ii) the submission of TSO mFRR needs to the mFRR Platform takes place at least 10 min before the beginning of the corresponding MTU, which is the common case in European markets [33], iii) the Gate Closure Time for the notification of the cleared balancing offers by the TSO to the IEVA is defined 10 min before the beginning of each MTU, iv) the prices for the mFRR Upward/Downward

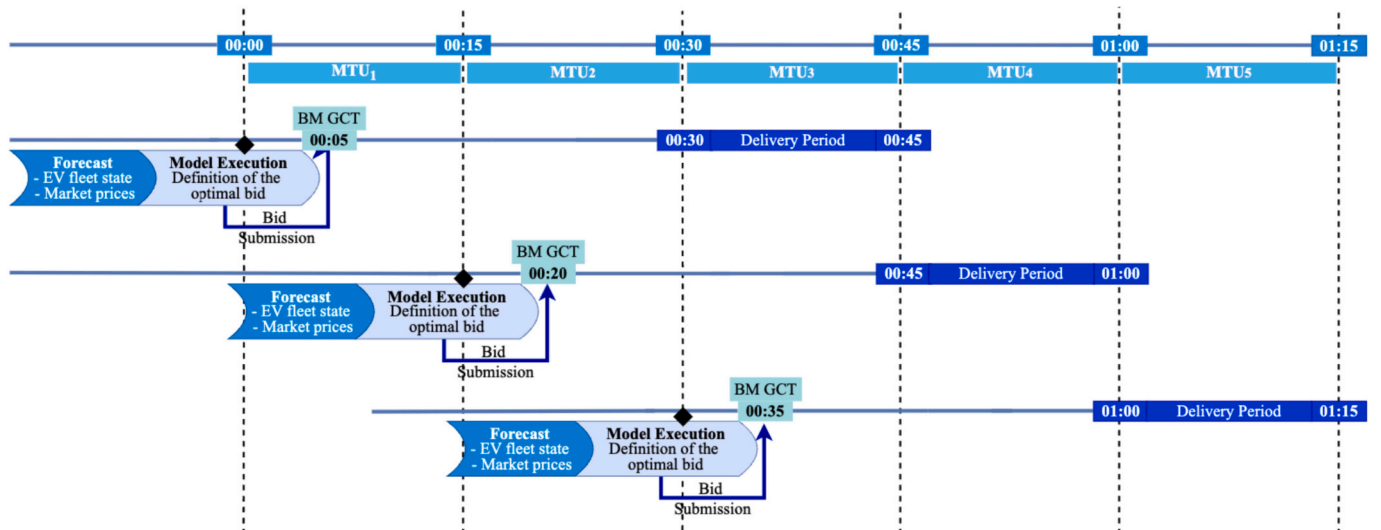


Fig. 1. The BM timeline and the relevant IEVA actions for rolling operation for three consecutive MTUs.

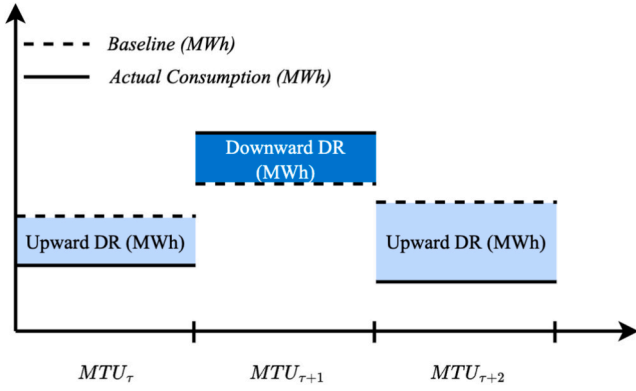


Fig. 2. The provided DR (i.e., upward/downward mFRR) equals the difference between the baseline and the actual EV fleet consumption.

Balancing Energy ($BE^{up/dn}$) product are based on marginal pricing (pay-as-cleared), v) the IEVA is remunerated when providing BE^{up} during positive mFRR upward ($mFRR^{up}$) prices and BE^{dn} during negative mFRR downward ($mFRR^{dn}$) prices.

2.2. Independent EV aggregator

The EV charging ecosystem is complex, mainly due to the involvement of many different parties and organizations that must interoperate effectively [38]. The IEVA is responsible for pooling the DR resources of multiple consumers and offering them as a single portfolio to the flexibility markets. The IEVA neither supplies energy to end users nor owns the charging infrastructure; however, it does employ information-and-communication technologies to make real-time decisions for optimal participation in the balancing market. Because EV users authorise the IEVA to manage charging remotely, a contractual relationship [38] is required to govern access to static data (e.g., battery size), real-time data (e.g., EV State-of-Charge (SOC)) and the terms of control. Furthermore, the IEVA must utilize appropriate communication protocols (i.e., OCPP, ISO 15118), to ensure real time information exchange with the charging stations and the EVs and to send dispatch instructions in real time promptly. This study assumes that (i) EVs are not capable of discharging energy back to the grid, ii) no local renewable generation is installed at the charging stations, and (iii) the stations are not equipped with stationary energy storage systems.

Several real-world implementations demonstrate that aggregation of EVs for grid services is not only technically feasible but already in practice. For example, companies such as Jedlix [39] and ev.energy [40] aggregate thousands of EVs to optimize charging schedules, reduce emissions, and participate in demand response programs. The EDF Group's Dreev [41] has deployed vehicle-to-grid (V2G) projects across Europe, using aggregated EV fleets to provide ancillary services such as frequency regulation. Octopus Energy [42] in the UK uses dynamic pricing and smart tariffs to aggregate residential EV charging loads, contributing to grid flexibility and cost efficiency. Meanwhile, Green-Flux [43] offers a platform for managing and aggregating charging infrastructure, enabling scalable smart charging across networks. These industry cases support the viability and growing adoption of aggregation-based models, reinforcing the relevance and potential impact of the approach proposed in this work.

2.3. DR baseline methodologies

The DR provision is quantified by the difference between the actual consumption and the baseline for the respective MTU (Fig. 2) thus, a well-designed baseline methodology is important for the success of any

DR program, since it determines the DR performance [44].

Although, several baseline methodologies found in the literature the most widely used are i) the Meter Before Meter After (MBMA), ii) the High^{XoFY} and iii) the Declarative [8].

- I. MBMA: In the MBMA baseline methodology, the baseline equals the consumption measured during the MTU preceding the MTU when mFRR activation is requested. In case that the IEVA offers balancing energy for consecutive MTUs, the baseline is determined by the consumption prior to the first activation period, resulting in a constant baseline during those MTUs for the entire activation period [5].
- II. High^{XoFY}: The calculation of the High^{XoFY} baseline, is more computationally intensive and consist of an initial DR baseline determination followed by appropriate further adjustments for more accurate load estimation during historical DR events. More specifically, the methodology consists of two main steps. The first step determines an initial High^{XoFY} baseline, and the second step adjusts the initial High^{XoFY} based on close to real-time consumption. To determine the initial High^{XoFY}, the system operator analyzes historical consumption profiles from comparable days (e.g., weekdays, Saturdays, Sundays, and holidays) and selects the highest consumption values from the X highest-demand days within the most recent Y-day window - as described in [45]. The final High^{XoFY} baseline is obtained by adding a baseline adjustment to the initial value. This adjustment equals the difference between the average actual consumption and the average baseline over the most recent three-hour adjustment window.
- III. Declarative: Declarative baseline methodologies rely on the forecasted EV load curve by the IEVA, which submits the consumption estimation to the TSO, before BM GCT or before another predefined deadline. They are considered susceptible to manipulation, but this potential can be mitigated by requesting the IEVA to submit its baseline earlier than near real time, driving however an inevitable decrease in EV load curve forecasting accuracy [46]. In this paper Declarative^{DA} defines the case that the baseline must be submitted before the Day-Ahead Market (DAM) GCT and Declarative^{RT} the case that the baseline must be submitted in near real-time (e.g., few minutes before the beginning of an MTU). Declarative baselines in the independent aggregator framework are analogous to the “buy-your-baseline” approach described in [47] for the integrated supplier-aggregator model. As with “buy-your-baseline”, declarative baselines may represent either honest forecasts or strategically manipulated declarations.

Table 1 presents the different baseline methodologies that are implemented across selected EU countries [5]. Most countries offer multiple baseline methodologies per flexibility product, while the discussion for the baseline methodologies that are more suited for each flexibility product is still open [5,6].

Table 1
Different baseline methodologies across Europe.

Country	FCR	mFRR	aFRR
Belgium	MBMA	MBMA, High ^{XoFY}	Declarative
Finland	MBMA	MBMA	MBMA
Great Britain	MBMA, Declarative	High ^{XoFY} , Declarative	MBMA, Declarative, High ^{XoFY}
Netherlands	MBMA	–	Declarative
Switzerland	MBMA	MBMA, Declarative, High ^{XoFY}	MBMA, Declarative
Greece	–	MBMA, High ^{XoFY}	–
France	MBMA	MBMA, Declarative, High ^{XoFY}	–

2.4. Modification of EV users' charging cost

The charging cost faced by EV users participating in smart charging programs can change for two main reasons. First, modifying the charging profile may alter the charging cost. An EV user billed under a fixed tariff (€/kWh) sees no change, but one on a dynamic tariff (€/kWh) can pay more or less if energy is drawn at different times than in conventional charging. Ideally, smart charging should leave the retail bill unchanged. Second, the IEVA may share part of its market-participation revenues with users. Profit sharing is therefore a key challenge: the IEVA must allocate a substantial share of its earnings to EV users to motivate enrolment. Prior studies [48,49] show that electricity cost savings are the strongest incentive for joining smart charging schemes; hence, both the IEVA's financial performance and the magnitude of the cost savings are crucial for their widespread adoption.

3. Market participation framework

This section presents the mathematical formulation of the BM optimal bidding strategy for IEVAs which is formulated as a mixed-integer linear programming problem.

The IEVA, at the beginning of the current MTU τ , calculates the optimal balancing energy bidding quantities for the 2nd MTU ahead and beyond. The problem is executed on a rolling basis, with each subsequent execution starting from 2nd MTU ahead and expanding to a specified horizon, i.e. 96 MTUs ahead. Fig. 1 presents the BM timeline and the corresponding IEVA actions. The problem outputs are the optimal balancing energy bids for each MTU. Bids for MTU $_{\tau}$ and MTU $_{\tau+1}$ are known from the previous balancing market clearing loops (Fig. 1). The bid for MTU $_{\tau+2}$ is binding and matches the quantity that the IEVA submits to the BM at the current gate closure. Bids for MTU $_{\tau+3}$ and beyond are advisory decisions since they will be recalculated in subsequent problem runs.

Perfect forecasts for EV fleet consumption and market prices are assumed. These, combined with the assumption of full mFRR bid clearance under a price-taker approach and the rolling model execution, enable the baseline level for MTU $_{\tau+2}$ to be known before the model execution for MTU $_{\tau+2}$. As a result, the baseline is a problem parameter. In the MBMA design, the baseline for MTU $_{\tau+2}$ is equal to the consumption during MTU $_{\tau+1}$ if the IEVA does not provide flexibility during MTU $_{\tau+1}$, or it is equal to the baseline of MTU $_{\tau+1}$ if the IEVA provides flexibility during MTU $_{\tau+1}$, based on the results of previous model execution. The High^{XoFy} is calculated using historical consumption data from preceding days, with adjustments based on consumption observed prior to real-time. The Declarative^{DA} and Declarative^{RT} values are computed by the IEVA based on expected consumption and submitted to the TSO one day and 25 min before real-time, respectively. For MTU $_{\tau+3}$ and beyond, the baseline is set equal to the baseline for MTU $_{\tau+2}$. However, this decision does not affect the IEVA's bidding strategy for MTU $_{\tau+2}$, as MTUs from $\tau + 3$ onward are not included in the objective. Instead, they are accounted for in constraints to ensure proper EV charging scheduling. Results for MTUs $\tau + 3$ and beyond are considered advisory and are recalculated during subsequent model executions. Given the above analysis, in the developed model the set of time intervals expands from MTU $_{\tau+2}$ to MTU $_{\tau+96}$. We let $t = 1$ refer to MTU $_{\tau+2}$, thus $t \in T$ implies that $t \in [1, 94]$. In addition, the set I (indexed by i) represents the electric vehicles managed by the aggregator.

The objective function (1) aims to maximize IEVA's revenues from participation in the BM, for MTU $_{\tau+2}$. Eq. (2) represents the energy deviation de_t (MWh) between the EV fleet's energy consumption e_t^{RT} (MWh) and the known baseline BL_t (MWh), (3) establishes that the energy deviation de_t equals the upward or downward balancing energy $e_t^{be,up}$ or $e_t^{be,dn}$ (MWh). Eq. (4) calculates the total energy consumption of the EV fleet by summing the individual consumption of each EV, $e_{i,t}$, (kWh). Constraint (5) defines the upper and lower bounds for the total

$e_t^{be,up}$ offered by the IEVA, with the maximum $e_t^{be,up}$ being provided when the EV fleet's consumption is reduced from the baseline to zero. Binary variable $u_t^{be,up}$ is used to model the constraint. Constraint (6) uses the max/(min) charging power $P_i^{ch,max/(min)}$ (kW), to set the bounds for the total $e_t^{be,dn}$ offered in each time interval t (i.e., with the MTU duration ΔT equal to 0.25 h). $PLUG_{i,t}$ is a parameter which equals 0 for the MTUs that the i -th EV is disconnected and 1 for the MTUs that the i -th EV is connected. Eq. (7) establishes the upper and lower bounds for the consumption $e_{i,t}$ of each connected EV and (8) calculates the State-of-Charge (SOC) trajectory for each EV, $soc_{i,t}$ (p.u.), which is equal with the SOC of the previous MTU and the energy consumed during that MTU considering EV battery capacity BAT_i^{Cap} (kWh) and charging efficiency N_i^{ch} (p.u.). Eqs. (9) and (10) specify the initial SOC, SOC_i^{level} (p.u.), and desired SOC at departure, SOC_i^{dep} (p.u.), for each EV. Constraint (11) uses the binary variables $u_t^{be,up}$ and $u_t^{be,dn}$ to ensure that balancing energy in each MTU can only be either upward or downward and (12) dictates that the IEVA can only provide balancing energy when the direction and relevant mFRR prices, $\Lambda_t^{be,up}$ or $\Lambda_t^{be,dn}$ (€/MWh) result in a payment from the TSO to the IEVA (i.e., mFRR^{up} when $\Lambda_t^{be,up}$ is positive, and mFRR^{dn} when $\Lambda_t^{be,dn}$ is negative).

$$\text{Max } (\Lambda_t^{BE,up} \cdot e_t^{be,up} - \Lambda_t^{BE,dn} \cdot e_t^{be,dn})_{t=1} \quad (1)$$

s.t.

$$de_t = e_t^{RT} - BL_t, \forall t \in T \quad (2)$$

$$de_t = e_t^{be,dn} - e_t^{be,up}, \forall t \in T \quad (3)$$

$$e_t^{RT} = 0.001 \cdot \sum_i e_{i,t}, \forall t \in T \quad (4)$$

$$0 \leq e_t^{be,up} \leq u_t^{be,up} \cdot BL_t, \forall t \in T \quad (5)$$

$$u_t^{be,dn} \cdot 0.001 \cdot \sum_i [P_i^{ch,min} \cdot PLUG_{i,t} \cdot \Delta T] \leq e_t^{be,dn} \leq u_t^{be,dn} \cdot 0.001 \cdot \sum_i [P_i^{ch,max} \cdot PLUG_{i,t} \cdot \Delta T] - BL_t, t \in T \quad (6)$$

$$P_i^{ch,min} \cdot PLUG_{i,t} \cdot \Delta T \leq e_{i,t} \leq P_i^{ch,max} \cdot PLUG_{i,t} \cdot \Delta T, \forall i \in I, t \in T \quad (7)$$

$$soc_{i,t+1} = soc_{i,t} + \frac{e_{i,t}}{BAT_i^{Cap}} \cdot N_i^{ch}, \forall i \in I, t \in T \quad (8)$$

$$soc_{i,t=1} = SOC_i^{level}, \forall i \in I \quad (9)$$

$$soc_{i,t=T_{dep}} = SOC_i^{dep}, \forall i \in I \quad (10)$$

$$0 \leq u_t^{be,up} + u_t^{be,dn} \leq 1, \forall t \in T \quad (11)$$

$$\Lambda_t^{be,up} \cdot e_t^{be,up} - \Lambda_t^{be,dn} \cdot e_t^{be,dn} \geq 0, \forall t \in T \quad (12)$$

4. Case study Design

The developed BM participation framework for the IEVA is evaluated across sixteen (16) scenarios. The case study results demonstrate the framework's effectiveness in modelling the aggregator's bidding performance and present innovative insights into the impact of the applied baseline design on market participation, flexibility provision and profit outcomes.

4.1. Scenarios set-up

The sixteen (16) scenarios are derived by combining:

- I. Two (2) real-world datasets of EV fleet portfolios

- II. Actual market clearing price data from two (2) wholesale electricity markets
- III. Four (4) demand response baseline designs

This approach provides a comprehensive analysis of IEVA's participation in BM, considering varying balancing market prices and patterns, EV fleet sizes and charging profiles.

4.2. Case study duration

The developed framework is evaluated for the entirety of 2021, with a 15-min time resolution, using a rolling simulation approach, i.e., the outputs of one run serve as inputs for the next one. Consequently, numeric results have been generated for 35,040 MTUs (i.e., 365 days * 96 intervals per day) across sixteen (16) scenarios.

4.3. Preparation of key parameters

The two historical, real-world EV fleet datasets are provided by the City of Boulder [50] (hereafter, Boulder), and by London Borough of Barnet [51] (hereafter, Barnet) and they include 9937 and 4470 charging sessions for 2021, respectively. They have similar average annual idle times (i.e., 67.9 % and 68.2 %), but differ in daily charging patterns. In the Boulder EV fleet case, the IEVA manages 102 charging points with a total capacity of 714 kW, while in the Barnet EV fleet case the IEVA manages 81 charging points with a total capacity of 567 kW. Fig. 3 presents the average hourly charging consumption for the two EV datasets. Boulder's peak hours are between 09:00 and 15:00 (i.e., "charge at work" profile) while Barnet's peak hours are between 17:00 and 20:00 (i.e., "charge at home" profile). Their datasets include records of EV sessions, i.e., connection and disconnection time, charging duration, total energy consumed, and charger id. The EV charging stations belong to the same bidding zone and are connected to the low voltage level of the distribution grid.

For the case study perfect EV fleet forecast is assumed, thus the values for T_i^{arr} and T_i^{dep} correspond to the actual times of each charging session, while E_i denotes the actual consumption during each charging event. $P_i^{ch,min}$ is set to zero, whereas the maximum charging rate, $P_i^{ch,max}$, is considered the estimated EV charger's maximum rate (i.e., 7, 11 or 22 kW for Level 2 EV chargers) by evaluating the charging time and the charging energy of each charging session. N_i^{ch} is set to 0.95 [52], and BAT_i^{Cap} ranges from 50 to 100kWh [53]. The initial SOC_i^{arr} is calculated using eq. (13) where E_i represents the actual energy consumption during each EV charging session. After each model execution SOC_i^{arr} is updated to reflect the new SOC_i^{level} .

$$SOC_i^{arr} = SOC_i^{dep} - E_i \cdot \frac{N_i^{ch}}{BAT_i^{Cap}}, \forall i \in I \quad (13)$$

The case study utilizes actual 2021 wholesale electricity market clearing prices from France and Greece. The data were retrieved from [54] and include mFRR^{up/dn} prices (15-min resolution), DAM Clearing prices (60-min resolution) and Imbalance Settlement Clearing prices (15-min resolution). The French and Greek markets were selected due to the different imbalance settlement pricing mechanisms. The French market uses dual imbalance pricing, while the Greek market single imbalance pricing. Fig. 4 presents the average hourly mFRR^{up/dn} prices for both markets, for 2021, indicating that the Greek market exhibits a larger price spread between mFRR^{up} and mFRR^{dn} prices compared to the French market.

The combination of the two EV fleet scenarios (i.e., Boulder, Barnet) and the two electricity market scenarios (i.e., France, Greece) results in four distinct scenarios, labelled as: a) France^{Barnet}, b) France^{Boulder}, c) Greece^{Barnet}, d) Greece^{Boulder}. These four scenarios are evaluated across the four baseline designs outlined in section 2.3, i.e., MBMA, High^{XofY}, Declarative^{DA}, Declarative^{RT}.

The above data are fed into the optimization model which is solved sequentially for each MTU in 2021. A perfect forecast of balancing

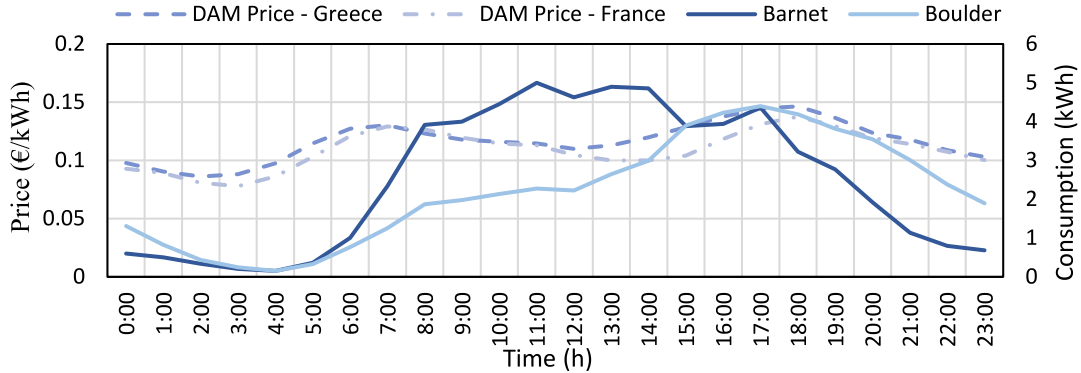


Fig. 3. Average hourly charging consumption profiles for the two EV fleets and mean hourly DAM clearing prices for the two electricity markets, for the year 2021.

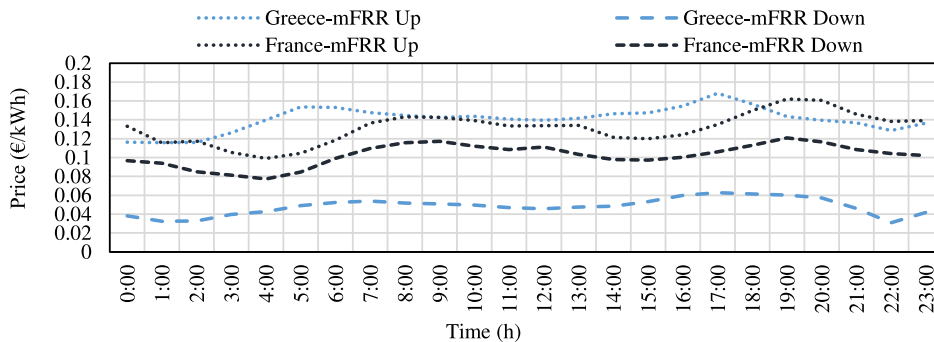


Fig. 4. Average hourly mFRRup/dn price profiles for the French and the Greek markets, for the year 2021.

market prices is assumed, meaning that the actual $mFRR^{up/dn}$ are used as input for the model. In addition, the perfect EV fleet forecast assumption leads to zero energy deviations between the dispatch instructions and the actual balancing energy delivered. These deliberate perfect-foresight assumptions keep the analysis focused on the impact of the work's key variable, the baseline design, enabling clearer comparison of economic outcomes across different EV fleets and markets.

5. Results analysis and discussion

This section presents the numerical results from the one-year long rolling simulation across the examined scenarios. The charging behavior of the EV fleets under smart charging is presented, followed by a detailed economic evaluation.

5.1. Fleets' charging behavior under smart charging

The IEVA's control over the charging process alters the initial charging schedule for each EV. Fig. 5 illustrates the plug-in times and charging durations for both direct and smart charging cases in the France^{Boulder} scenario. In the direct charging case, EVs begin charging immediately upon being plugged in. In contrast, in the smart charging approach the EVs charge according to the proposed strategy resulting in a longer charging duration. However, despite the extended charging time, the desired SoC is reached before T_i^{dep} in every case.

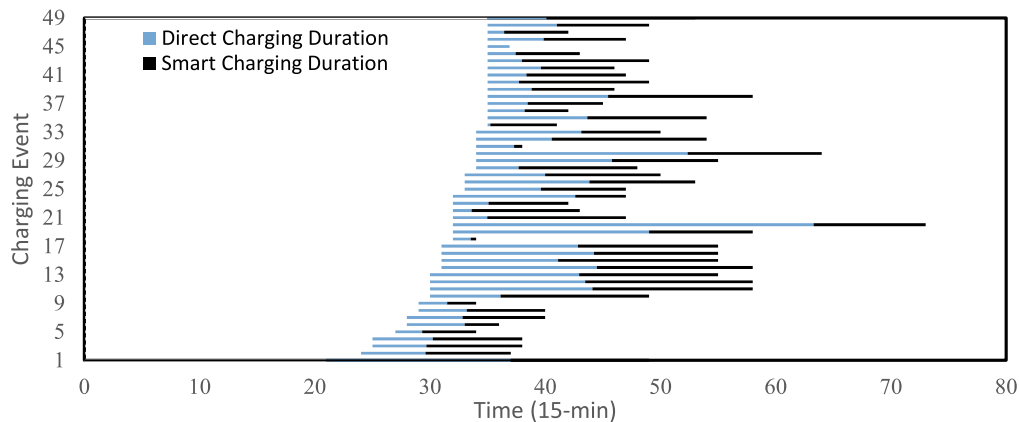


Fig. 5. Charging duration for direct and smart charging approaches for the first 50 charging events of the year 2021 for the France^{Boulder} scenario.

Table 2

Annual (2021) economic results for the sixteen scenarios for the one year rolling simulation.

C1	C2	C3	C4	C5	C6	C7
EV fleet & market Scenario	Baseline methodology	Annual Charging Energy (MWh)	Annual Provided BE ^{up} (MWh)	Annual Provided BE ^{dn} (MWh)	IEVA annual revenues (10 ³ €)	IEVA annual revenues (€/kWh)
France ^{Barnet}	MBMA	76.8	231.8	0.365	33.52	0.43
	High ^{XoFY}		102.2	0.832	13.15	0.17
	Declarative ^{DA}		62.7	0.663	10.06	0.13
	Declarative ^{RT}		205.3	–	31.00	0.40
	MBMA		179.2	0.106	23.36	0.26
France ^{Boulder}	High ^{XoFY}	87.9	75.2	0.452	11.34	0.12
	Declarative ^{DA}		64.2	0.251	9.44	0.10
	Declarative ^{RT}		124.2	–	18.46	0.20
	MBMA		217.9	1.5	32.26	0.41
	High ^{XoFY}		106.7	10.5	14.91	0.19
Greece ^{Barnet}	Declarative ^{DA}	76.8	61.9	12.9	9.52	0.12
	Declarative ^{RT}		188.5	–	29.00	0.37
	MBMA		165.8	0.65	25.39	0.28
	High ^{XoFY}		79.5	5.1	12.68	0.14
	Declarative ^{DA}		59.9	4.3	9.63	0.10
Greece ^{Boulder}	Declarative ^{RT}	87.9	114.5	–	19.13	0.21

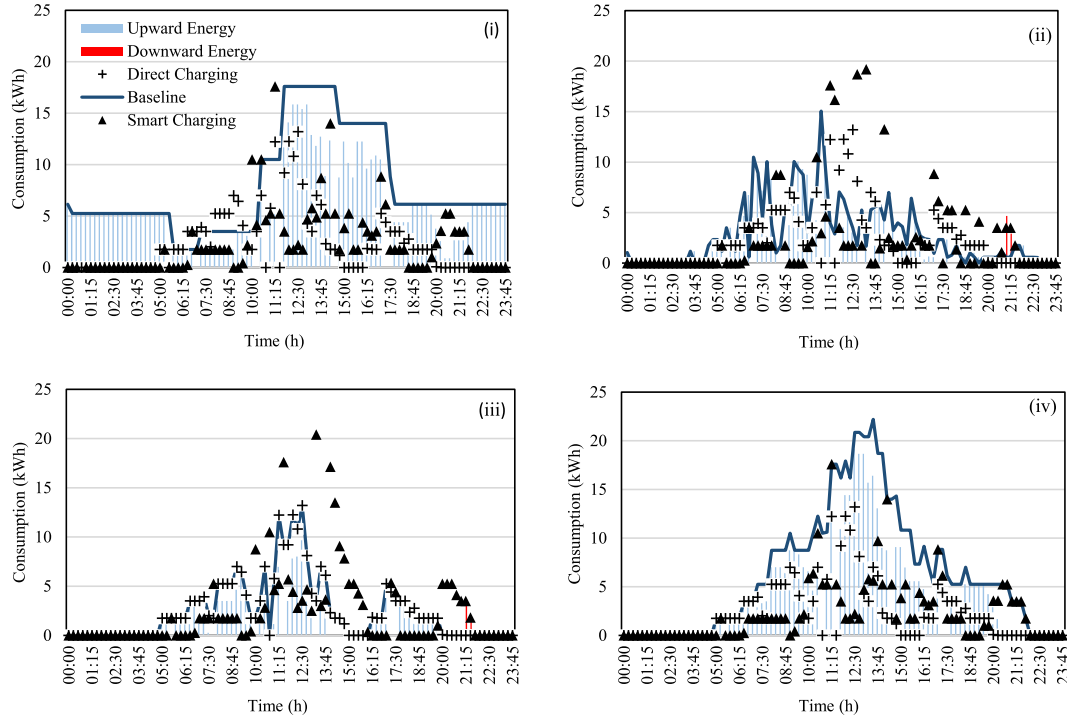


Fig. 6. Baselines, BE^{up} , BE^{dn} and charging trajectories under direct and smart charging for the (i) MBMA, (ii) $High^{XofY}$, (iii) $Declarative^{DA}$ and (iv) $Declarative^{RT}$ baseline, for France^{Barnet} scenario, for 09/05/2021.

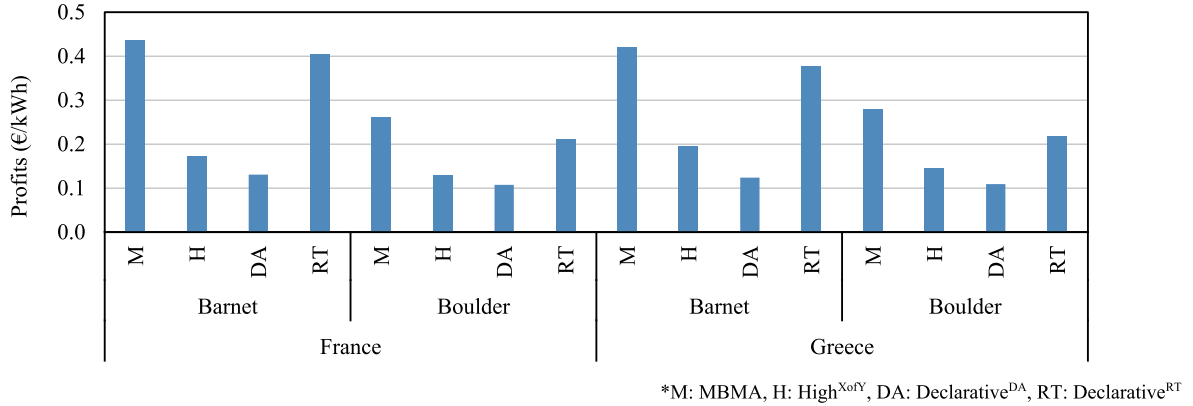


Fig. 7. The average annual revenues in €/kWh, for each scenario and baseline methodology. *M: MBMA, H: $High^{XofY}$, DA: $Declarative^{DA}$, RT: $Declarative^{RT}$

Consequently, when the IEVA provides balancing energy for multiple consecutive MTUs, the baseline corresponds to the consumption of the last MTU that both BE^{up} and BE^{dn} are zero. This can lead to manipulation of the baseline level, as presented in Fig. 6 (i), where the baseline equals the consumption before the first of those MTUs. This approach allows the IEVA to strategically increase consumption before specific periods and provide then BE^{up} without reducing the charging rate of the EV fleet. In the case of the $High^{XofY}$ baseline design (Fig. 6 (ii)), the baseline is calculated using historical consumption data during periods without DR events. In this design the IEVA, when possible, can dispatch the EV fleet consumption level to deviate from the calculated baseline and offer the deviation as BE^{up} or BE^{dn} . Furthermore, when using the $Declarative^{DA}$ baseline methodology (Fig. 6 (iii)) the IEVA declares the expected baseline the day before the delivery period. By assuming perfect EV fleet forecast, the baseline matches the actual consumption under direct charging. Finally, in the case of the $Declarative^{RT}$ baseline design (Fig. 6 (iv)) the baseline is declared before the balancing offers' submission and corresponds to a close to real-time forecast of the EV fleet consumption.

When the aggregator submits a BE^{up} offer, the remuneration is based on the difference between the baseline and the final consumption. Consequently, in $Declarative^{RT}$ as well as MBMA baseline designs the aggregator might artificially inflate the expected consumption by submitting a higher baseline. The opposite strategy applies for the BE^{dn} offers.

Provision of downward balancing services is also possible when the consumption level deviates from the baseline and $mFRR^{dn}$ prices are negative. In such cases, the IEVA is compensated for increasing the EV fleet consumption level above the baseline level. This is depicted in Fig. 6 (iii) and Fig. 6 (iv), where the aggregator provides BE^{dn} for a few MTUs per scenario. The frequency of $mFRR^{dn}$ activations is related to the frequency of negative $mFRR^{dn}$ prices. In 2021, the prices for BE^{dn} were negative for 1 % of the MTUs in the French balancing market and for 8 % of the MTUs in the Greek balancing market. Consequently, the IEVA in the Greek market tends to offer more BE^{dn} for both EV fleet cases compared to the counterpart in the French market. The annual BE^{dn} amount is presented in Table 2 (column C5) and is significantly lower than the annual BE^{up} amount.

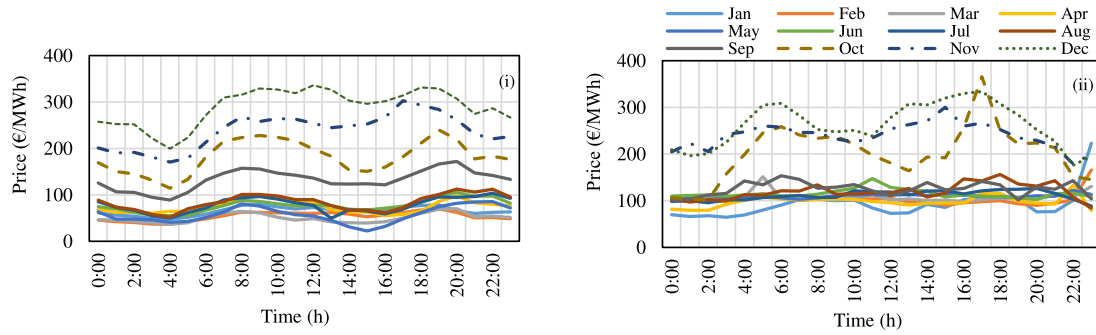


Fig. 8. The hourly average clearing prices for the mFRRUp per month, for France (i) and Greece (ii), for 2021.

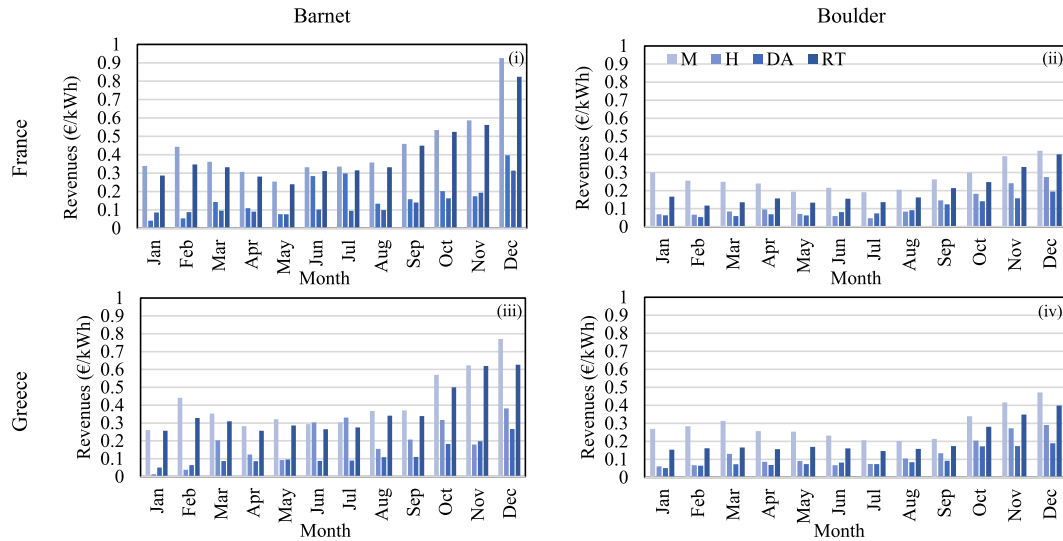


Fig. 9. The average monthly revenues per kWh charged, for (i) France^{Barnet}, (ii) France^{Boulder}, (iii) Greece^{Barnet}, (iv) Greece^{Boulder} scenarios and for each applied baseline methodology. *M: MBMA, H: High^{XoFY}, DA: Declarative^{DA}, RT: Declarative^{RT}.

Table 3

Annual economic impact on energy supplier's portfolio for the sixteen scenarios for the one year rolling simulation.

C1	C2	C3	C4	C5	C6
EV fleet & market Scenario	Baseline methodology	Annual Charging Energy (MWh)	Supplier annual cost under direct charging (10 ³ €)	Supplier annual cost under smart charging (10 ³ €)	Difference (%)
France ^{Barnet}	MBMA	76.8	9.6	9.86	2.7 %
	High ^{XoFY}			9.94	3.5 %
	Declarative ^{DA}			9.68	0.8 %
	Declarative ^{RT}			9.86	2.7 %
France ^{Boulder}	MBMA	87.9	11.4	12.0	5.2 %
	High ^{XoFY}			12.3	7.9 %
	Declarative ^{DA}			12.2	7.0 %
	Declarative ^{RT}			12.1	6.1 %
Greece ^{Barnet}	MBMA	76.8	10.8	10.3	-4.6 %
	High ^{XoFY}			10.5	-2.8 %
	Declarative ^{DA}			10.1	-6.5 %
	Declarative ^{RT}			10.1	-6.5 %
Greece ^{Boulder}	MBMA	87.9	11.8	11.6	-0.8 %
	High ^{XoFY}			11.6	-1.7 %
	Declarative ^{DA}			11.7	-0.8 %
	Declarative ^{RT}			11.6	-1.7 %

5.3. Economic results evaluation

5.3.1. Independent EV aggregator revenues

The expected revenue generation is a key aspect when the IEVA designs its business model [38]. In this section, the annual and monthly analysis of the average expected revenues of the IEVA from BM participation, across all sixteen scenarios is presented. Section 5.3.1.1 presents

the annual revenue analysis, while section 5.3.3.2 presents the monthly revenue analysis, for 2021.

5.3.1.1. Annual revenue analysis. Table 2 also presents the economic results for the sixteen scenarios, over the one-year rolling simulation. Column C6 presents the annual expected revenues for the IEVA in each scenario in euros (€), while Column C7 provides the revenue in € per

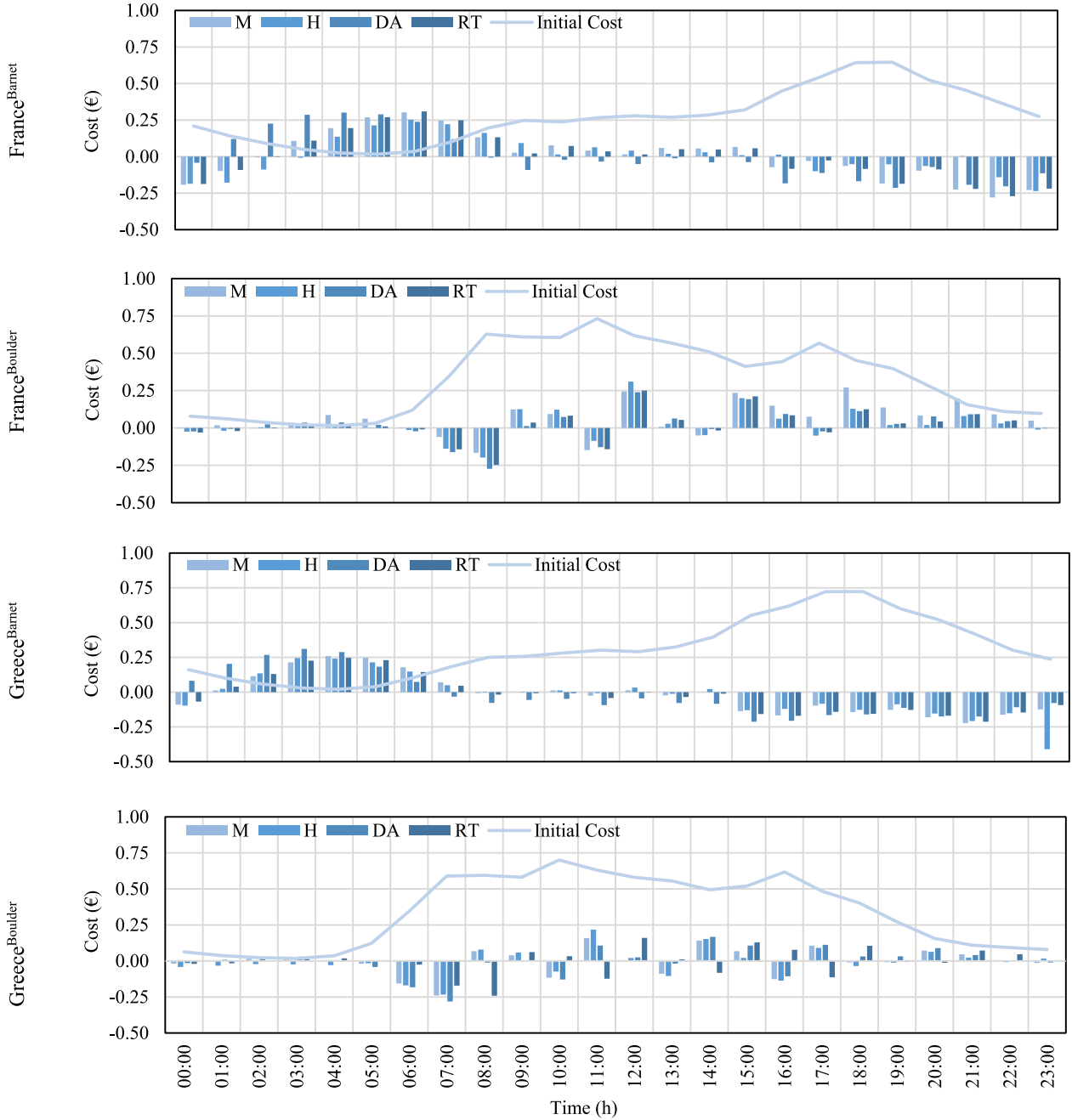


Fig. 10. The initial hourly average supplier's procurement cost and the average cost increase/decrease under smart charging for all scenarios and MBMA, High^{XofY}, Declarative^{DA} and Declarative^{RT} baselines, for 2021.

kWh charged (€/kWh), which is presented also in Fig. 7. The revenues from BM participation are significant, providing a strong margin to the IEVA to establish a smart charging business. Indicatively, in France^{Barnet} scenario, the revenues per kWh charged are as follows: 0.43€/kWh for the MBMA baseline, 0.40€/kWh for the Declarative^{RT}, 0.17€/kWh for High^{XofY} and 0.13€/kWh for the Declarative^{DA}. Similar results are observed across the other EV fleet and market scenarios, with MBMA being the most profitable baseline methodology, followed by Declarative^{RT}, High^{XofY} and Declarative^{DA}.

5.3.1.2. Monthly revenue analysis. Fig. 8 presents the hourly average clearing prices for the mFRR^{up} per month, in France and Greece. The average mFRR^{up} price for both markets rises significantly in the fourth quarter of 2021 due to changes in market conditions.

Fig. 9 presents the average monthly revenue per kWh charged, across all sixteen scenarios. The MBMA baseline methodology consistently generates the highest revenue, for every month and scenario. The increased revenues in the last quarter of 2021 across all baseline methodologies result from the respective rise in mFRR^{up} price (Fig. 8).

5.3.2. IEVA's impact on supplier's energy procurement cost

The supplier's procurement cost represents the expense incurred in purchasing energy from the wholesale energy market to charge the EV fleet. The direct cost ($ES_t^{\text{direct cost}}$) refers to the direct charging scenario, without IEVA, and it is calculated in (14), considering the hourly energy consumption, E_t^{RT} , and the corresponding DAM Clearing price, Λ_t^{DAM} . Under direct charging, it is assumed that the consumption matches the energy purchased from the market (E_t^{MRKT}) resulting in no energy

Table 4

Annual (2021) economic results for the charging cost per kWh paid by the EV user to the Supplier, across all sixteen scenarios.

C1	C2	C3	C4	C5	C6
EV fleet & market Scenario	Baseline methodology	Fixed tariff - Direct charging (€/kWh)	Dynamic tariff (weighted) - Direct charging (€/kWh)	Dynamic tariff (weighted) - Smart charging (€/kWh)	Dynamic tariff (weighted) - Charging cost deviation (smart vs direct)
France ^{Barnet}	MBMA	0.109	0.126	0.121	-4 %
	High ^{XoFY}			0.122	-3 %
	Declarative ^{DA}			0.118	-6 %
	Declarative ^{RT}			0.122	-3 %
France ^{Boulder}	MBMA	0.131	0.131	0.130	-1 %
	High ^{XoFY}			0.130	-1 %
	Declarative ^{DA}			0.130	-1 %
	Declarative ^{RT}			0.130	-1 %
Greece ^{Barnet}	MBMA	0.116	0.142	0.137	-4 %
	High ^{XoFY}			0.137	-4 %
	Declarative ^{DA}			0.132	-7 %
	Declarative ^{RT}			0.136	-4 %
Greece ^{Boulder}	MBMA	0.134	0.134	0.134	0 %
	High ^{XoFY}			0.134	0 %
	Declarative ^{DA}			0.133	-1 %
	Declarative ^{RT}			0.134	0 %

imbalance. The supplier's procurement cost under smart charging ($ES_t^{\text{smart cost}}$) is calculated in (15), accounting for energy imbalances between E_t^{MRKT} and E_t^{RT} , due to the smart charging managed by the IEVA and the corresponding hourly imbalance price, Λ_t^{imb} .

$$ES_t^{\text{direct cost}} = \sum [E_t^{\text{MRKT}} \cdot \Lambda_t^{\text{DAM}}] \quad (14)$$

$$ES_t^{\text{smart cost}} = \sum [E_t^{\text{MRKT}} \cdot \Lambda_t^{\text{DAM}} - (E_t^{\text{MRKT}} - E_t^{\text{RT}}) \cdot \Lambda_t^{\text{imb}}] \quad (15)$$

The selected electricity markets use two different imbalance settlement mechanisms. The Greek electricity market follows a single imbalance settlement mechanism, where imbalances are settled at a single price, regardless of whether they arise from excess supply or excess demand. This pricing approach incentivizes generators and suppliers to help restore system imbalance. In contrast, the French electricity market follows a dual imbalance settlement mechanism, applying different prices for positive (excess supply) and negative (excess demand) imbalances. This approach incentivizes energy generators and suppliers to maintain their own balance [55].

Table 3 presents the economic impact of smart charging on the supplier's total annual cost. Column C4 presents the annual cost for the supplier under direct charging, while column C5 presents the annual cost considering the final charging schedule due to smart charging. Column C6 presents the difference between the initial and the final annual cost. Smart charging managed by the IEVA results in a decrease in the supplier's annual procurement cost in the Greek electricity market, but an increase in the French electricity market. The explanation for this result is the following: When the IEVA reduces the EV fleet charging compared to the energy purchased by the supplier in the market, E_t^{MRKT} , for a given MTU, an energy surplus (positive energy imbalance) is created for the supplier, during that same MTU. Conversely, when the IEVA increases EV fleet charging compared to E_t^{MRKT} , an energy deficit (negative energy imbalance) is created for the supplier. The supplier's final energy procurement cost varies based on the size of the imbalance (MWh) and the prices spread in the market. Typically, imbalance prices are positive, meaning that reducing consumption lowers the supplier's imbalance cost, while increasing consumption raises it. The overall final cost depends on the imbalance prices when consumption deviates from the E_t^{MRKT} . Fig. 10 presents the initial hourly average procurement cost for the supplier, as well as the average cost increase/decrease under smart charging for all scenarios and baseline methodologies. While the overall volumes (kWh) of the positive and the negative energy imbalances for the supplier are equal, the monetary value (€) of these imbalances differs, resulting in varying final annual costs for the supplier. In the Greek energy market, when energy consumption decreases,

imbalance prices tend to be higher than when consumption increases, resulting in a lower final procurement cost. In contrast, in the French market, when consumption is reduced, imbalance prices tend to be lower, resulting in higher final procurement cost. The difference between the $ES_{\text{cost}}^{\text{initial}}$ and $ES_{\text{cost}}^{\text{final}}$ across the applied baseline methodologies and scenarios ranges between -6.5 % and 7.9 %, indicating that in the long term the IEVA's BM participation impacts the suppliers' balancing position.

5.3.3. Modification of EV users' charging cost

Following the analysis of section 2.4 the EV user charging cost is evaluated for two different charging tariff structures, fixed and dynamic [56].

- Fixed tariff (€/MWh): This tariff remains constant throughout the case study, and it is set equal to the annual average DAM clearing price of each electricity market. For France, it is 0.109 €/kWh and for Greece, 0.116 €/kWh (Table 4, C3).
- Dynamic tariff (€/MWh): This tariff fluctuates hourly, matching the DAM clearing price (French or Greek). A smart meter is required for the application of the dynamic tariff. The weighted tariff for each scenario is presented in Table 4, C4.

Fig. 11 shows the hourly average DAM clearing price for each scenario, the hourly average annual direct EV charging profile for each EV fleet and the hourly average increase or decrease in the EV fleet consumption under smart charging, across all sixteen scenarios, for 2021. Under direct charging, the EV fleet of Barnet (charge at home) experiences peak demand during hours when the DAM clearing prices are high in both electricity markets. In contrast, EV fleet of Boulder (charging at work) reaches peak energy demand during the valley of the DAM clearing prices.

Under the smart charging scheme, EV consumption shifts to the following MTUs, since the IEVA primarily provides upward balancing energy. With the dynamic tariff, the new charging cost for the EV user is affected by the alignment between the DAM clearing price profile and the final EV charging profile. As explained in Fig. 11, for the case of the Barnet EV fleet (i.e., charge at home), peak demand under direct charging coincides with higher DAM clearing prices. Therefore, when the IEVA shifts this consumption to the following MTUs, where DAM clearing prices are lower, it results in a reduced charging cost for the EV user under smart charging (i.e., 4–7 % cost decrease, Table 4, column C6). In contrast, for the case of the Boulder EV fleet (charge at work), peak demand under direct charging aligns with lower DAM clearing prices. This keeps the charging cost relatively stable (i.e., less than 1 %

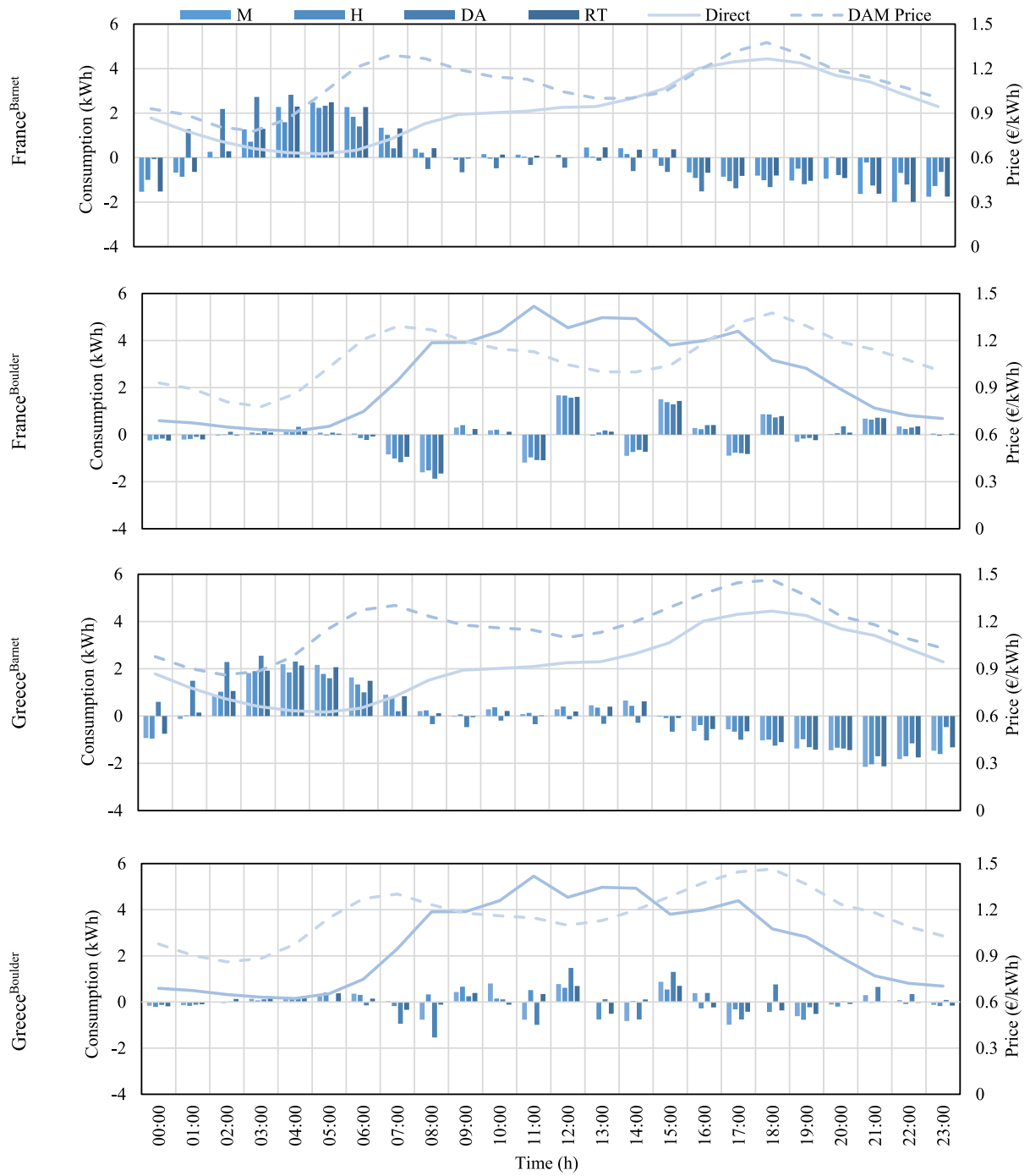


Fig. 11. The hourly average DAM clearing prices for France, the hourly average direct EV charging profile and the hourly average increase/decrease of EV fleet consumption under smart charging for all cases for MBMA, High^{xy}, Declarative^{DA} and Declarative^{RT} baselines, for 2021.

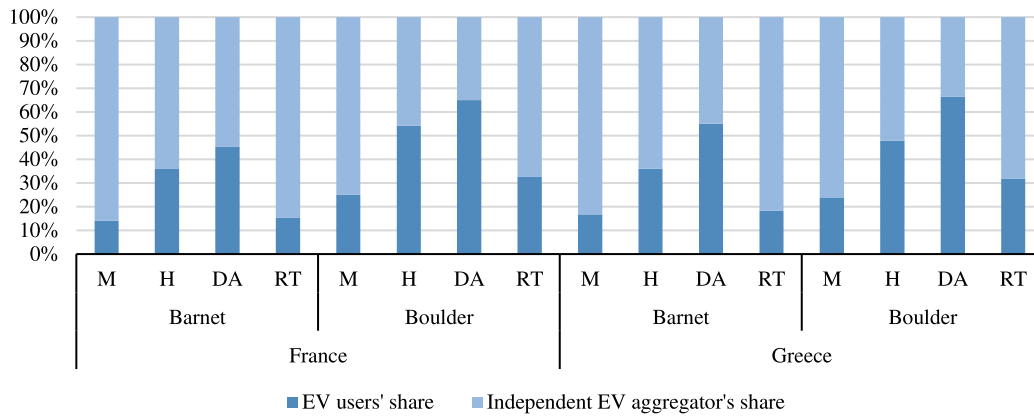


Fig. 12. The profit sharing between the independent EV aggregator and the EV users, to ensure 50 % charging cost reduction.

decrease, Table 4, Column C6) as the final EV charging profile under smart charging continues to align with the low DAM clearing prices.

Fig. 12 presents the average profit-sharing ratio between the IEVA and the EV users, when the IEVA incentivizes smart charging participation by offering a 50 % reduction in the users' final charging cost. As shown, the IEVA can provide this significant reduction by sharing 14–25 % of the profits under the MBMA baseline, 35–55 % under the High^{Xofy}, 45–66 % under the Declarative^{DA}, and 15–33 % under the Declarative^{RT}, depending on the EV fleets and electricity market scenarios. These results emphasize the potential of smart charging to lower EV users' charging costs, while also demonstrating the impact of baseline methodologies selection on both the IEVA's profits and EV users' charging cost savings.

6. Conclusion and policy implications

Well-designed policies for demand response mechanisms require efficient baseline designs that address the critical requirements of accuracy, fairness, simplicity and interoperability in quantifying flexibility provided by various loads. Although many baseline designs have been operationally evaluated for certain load types, their suitability for electric vehicles remains largely unexplored. In response, this paper proposes a comprehensive balancing market participation framework for an independent electric vehicle aggregator that manages electric vehicle fleet charging, a process commonly known as smart charging. The framework was tested across four baseline designs, using two electric vehicle fleet datasets and data from two European electricity markets, in year-long simulations, providing policy insights on the question: Are current demand response baseline designs suitable for effective demand response provision by electric vehicles?

We first examined this question from a business perspective by evaluating the aggregator's profitability from the balancing market participation as well as the charging cost savings potential for the EV users, assuming full aggregator's bids clearance under the price-taking approach. The numerical results indicate that the market participation revenues for workplace and residential charging are significant, ranging from 0.13€ to 0.43€ per kWh charged. The Meter-Before-Meter-After baseline design, which relies on consumption data from the latest 15-min period prior to the demand response event, consistently generates the highest revenue, followed closely by the Real-Time Declarative design, which is based on near real-time aggregator's declarations. In contrast, the High^{Xofy} baseline, which uses historical consumption adjusted by further modifications, and the Day-Ahead Declarative design, which relies on day-ahead consumption forecasts, lead to lower revenues. In addition, since discounts in charging fees are a crucial incentive for EV users to participate in smart charging programs [14,57] we considered a scenario in which the aggregator offers a 50 % discount on the users' initial charging cost. To achieve this, the aggregator shares

14–25 % of the profits under the Meter-Before-Meter-After baseline, 15–33 % under the Real-Time Declarative design, 35–55 % under the High^{Xofy}, and 45–66 % under the Day-Ahead Declarative design, depending on the fleet and market scenario. Thus, even in the least profitable scenarios, the revenue margins indicate promising potential for the independent aggregation business model.

From a policy perspective these novel results are enlightening. Higher profits are driven by larger volumes of flexibility provided, which vary across baseline designs. Specific designs, like High^{Xofy} and Day-Ahead Declarative, are less prone to manipulation, with flexibility volumes ranging between 75 and 130 % and 70–80 % of the charging energy, respectively. In contrast, designs such as Meter-Before-Meter-After and Real-Time Declarative are more susceptible to baseline manipulation, allowing the aggregator to provide flexibility volumes between 200 and 300 % and 150–300 % of the charging energy, respectively. In Meter-Before-Meter-After, an independent aggregator can strategically increase EV consumption ahead of selected periods and then offer upward balancing energy without reducing the fleet's charging rate. This is possible because, when balancing energy is supplied for several consecutive market time units, the baseline is set equal to the consumption recorded in the most recent unit for which no balancing energy was delivered. In Real-Time Declarative design the baseline is declared before the balancing offers' submission and the aggregator might artificially inflate the expected consumption by submitting a higher baseline. In both designs, "baseline leverage" occurs when flexibility is credited even though the charging level never actually deviates from the schedule, thereby artificially inflating the reported flexibility and the resulting profits.

These results suggest that baseline designs suited for curtailable loads (e.g., industries with flat or predictable consumption profiles) may not be appropriate for highly flexible and shiftable loads such as electric vehicles. Excessive baseline leverage can lead to unfair competition among flexibility providers and ultimately threaten the viability of the demand response market. Our findings contribute to the ongoing debate over the effectiveness of current baseline designs, which have been subject to criticism [58,59]. Consequently, it is crucial for system operators and policy makers to recognize the potential distortions imposed by different baseline designs and ensure that available baselines promote fair competition among demand response providers. Establishment of appropriate quality factors [60] to identify and address these distortions should be widely promoted. In addition, in markets where multiple baseline designs are available following the "no one-size-fits-all" principle [7] and flexibility providers can choose among them, policymakers should reconsider access rights for all types of loads to all types of baselines, to uphold the fairness criterion. Moreover, further research on emerging baseline designs, such as those that utilize control groups [15,58] or "engineered" approaches [15], is essential. However, these designs must also adhere to the simplicity and interoperability criteria.

Finally, as noted in the case study, the deliberate assumption of perfect forecasts for both the EV fleet state and the balancing market clearing prices keeps the analysis focused on the study's key variable, i. e. the baseline design, and enables clear comparison of economic outcomes across different fleets and markets. Therefore, the results serve as a benchmark for optimal market participation under perfect foresight. Imperfect forecasts, however, introduce two main risks. First, uncertainty in vehicles' arrival and departure times and state-of-charge can cause the independent aggregator to under-deliver the activated flexibility, potentially incurring non-compliance charges. An EV disconnection during upward mFRR delivery hopefully, has little adverse impact because it inherently lowers consumption, which supports the service. In contrast, a disconnection while providing downward mFRR can reduce consumption below the baseline and create an imbalance that is harder to correct. Notably, the case study shows that the aggregator supplies upward mFRR almost exclusively, so exposure to this risk is limited. Second, inaccurate price forecasts can lead to sub-optimal bid quantities, reducing revenues relative to the perfect forecast benchmark. These risks are nevertheless substantially mitigated by the balancing market rules that allow continuous bid revisions, enabling the independent aggregator to update its offers at each 15-min gate closure. As a result, forecast errors affect only the next market time unit and do not propagate through the entire bidding horizon. The quantitative impact of such imperfect forecast conditions will be investigated in future work.

CRedit authorship contribution statement

Konstantinos D. Afentoulis: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Stylianios I. Vagropoulos:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research project was supported by the Hellenic Foundation for Research and Innovation (H.F.R.I.) under the “2nd Call for H.F.R.I. Research Projects to support Post-Doctoral Researchers” (Project Number: 649, Project Name: Smart electric vehicles integration in electric energy systems)."

Data availability

Data will be made available on request.

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